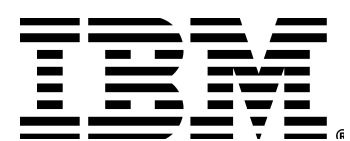


# Containers & OpenShift Workshop

Alexander Mamani - IBM  
Kate Queiroz - Red Hat  
Christian Zaccaria - Red Hat  
Winnie Imafidon - IBM



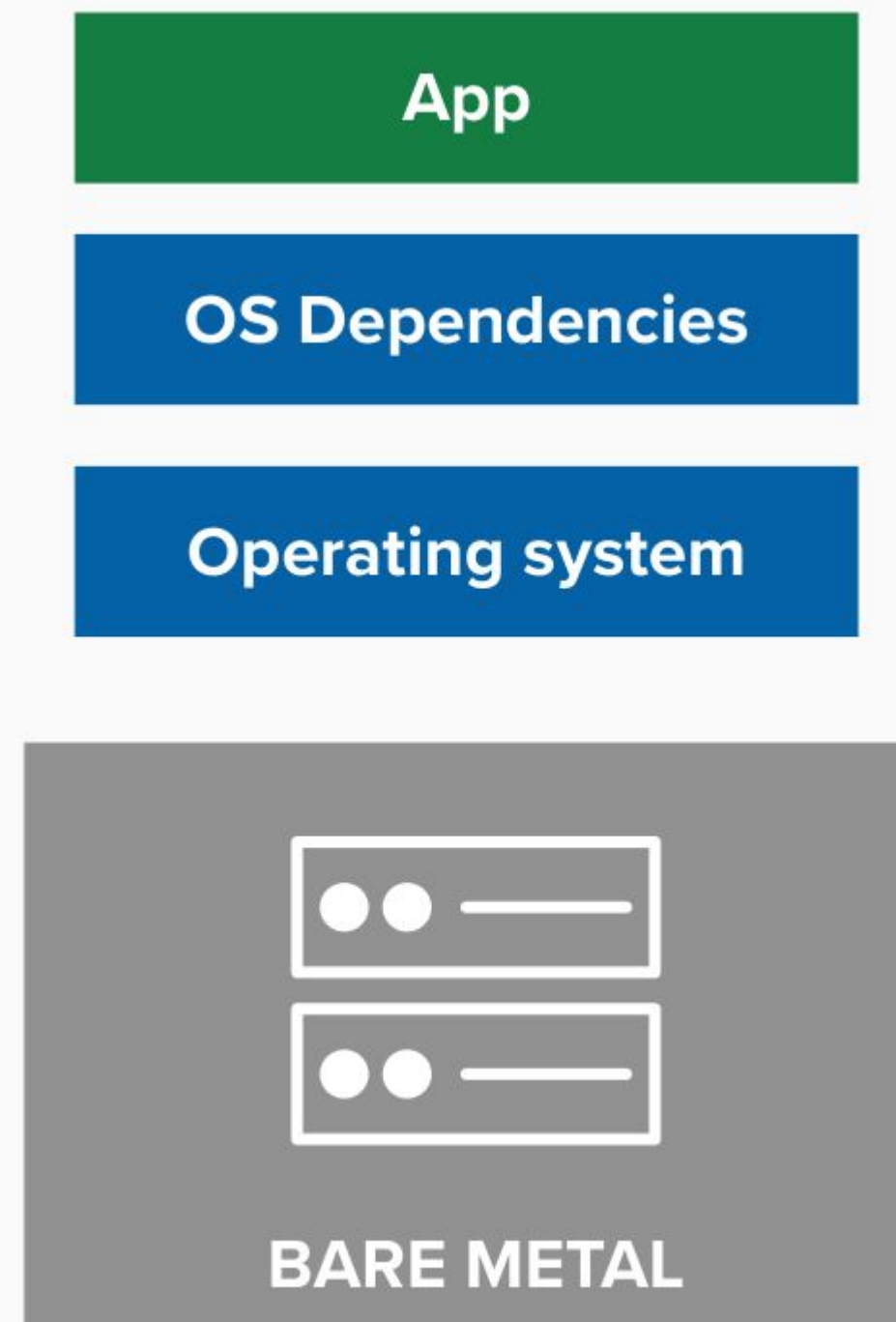
# A LITTLE BIT ABOUT VIRTUALIZATION



**Red Hat**

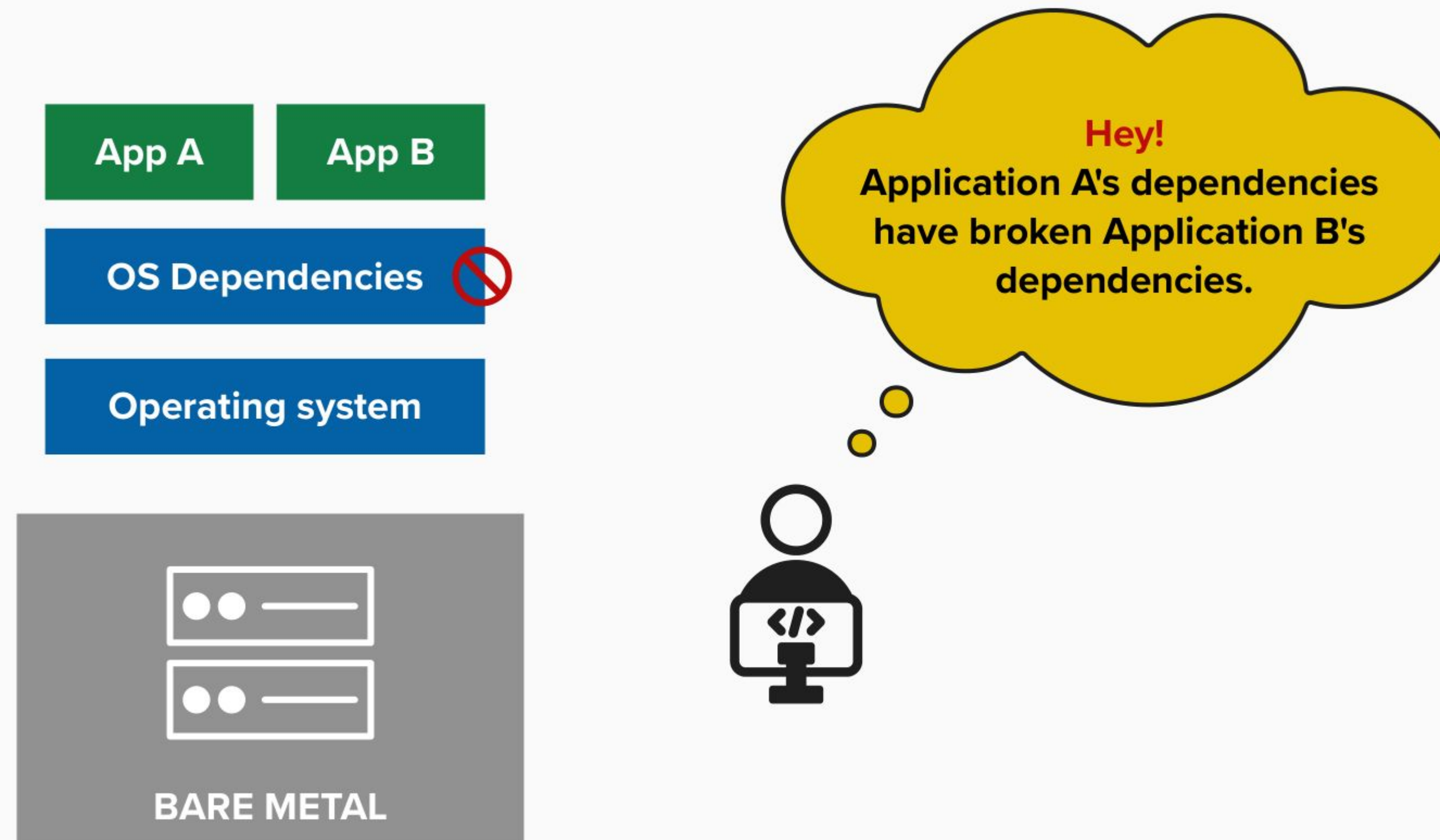


# 1. Traditional deployment



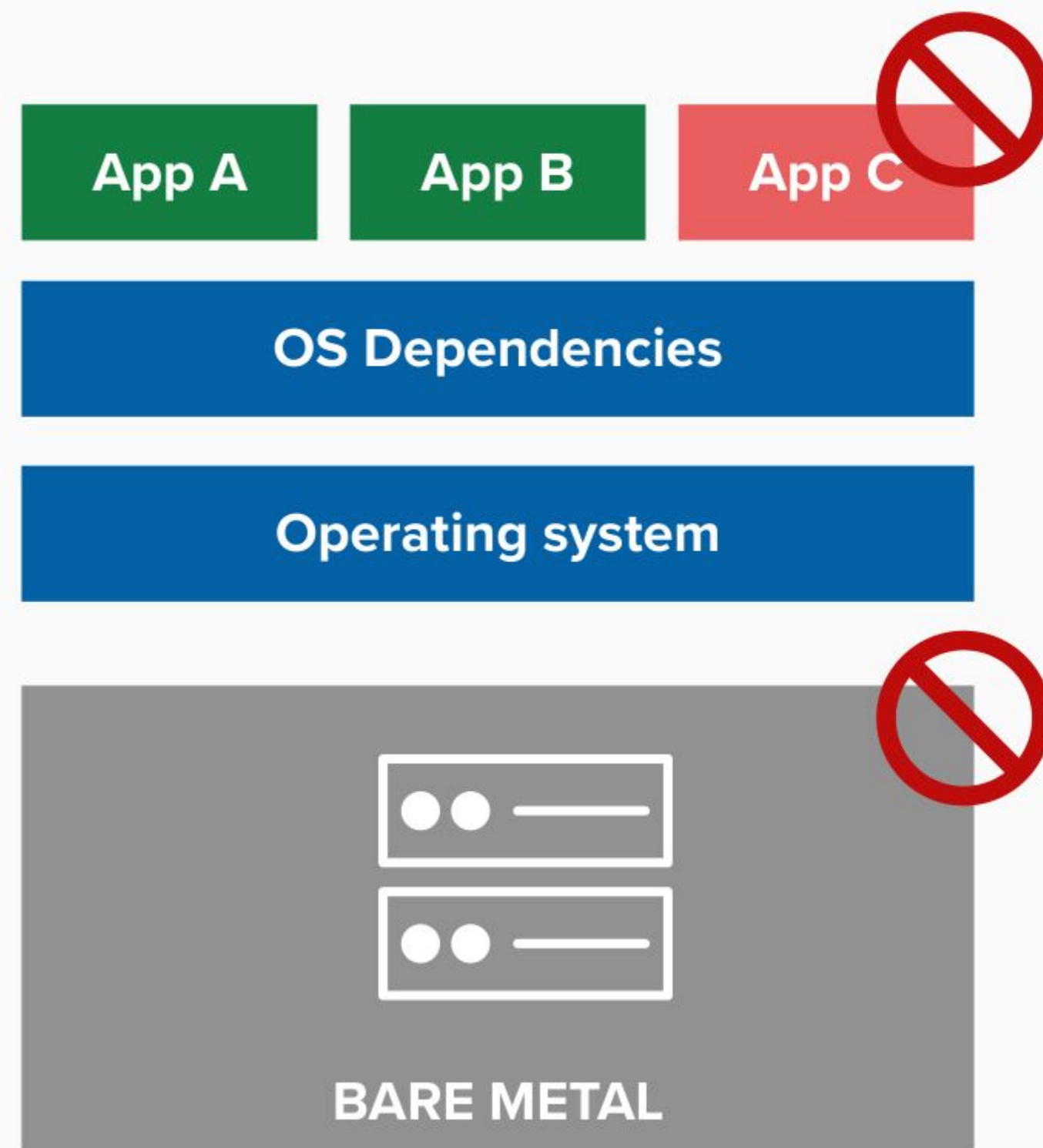


# 1. Dependency Management Challenge

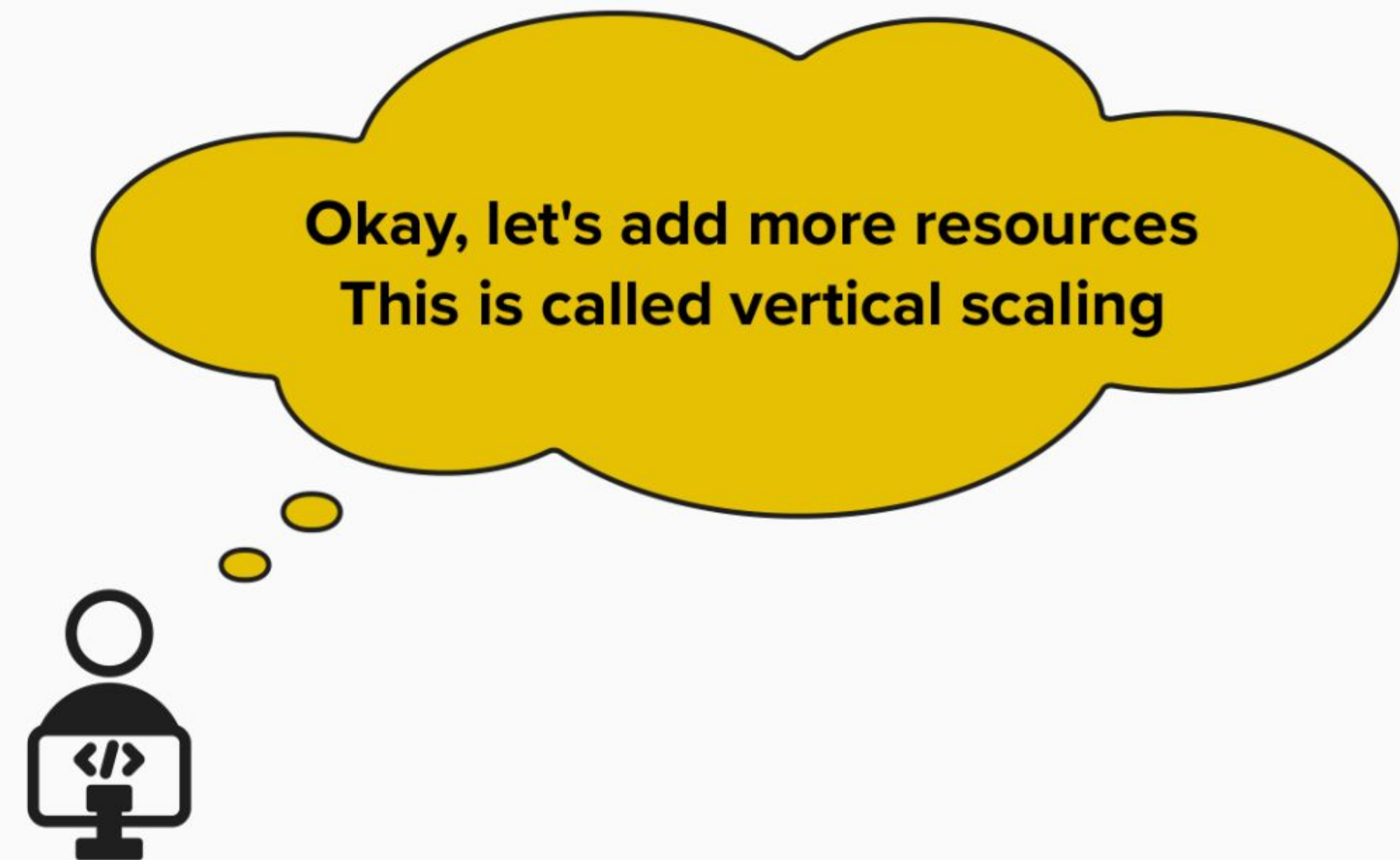
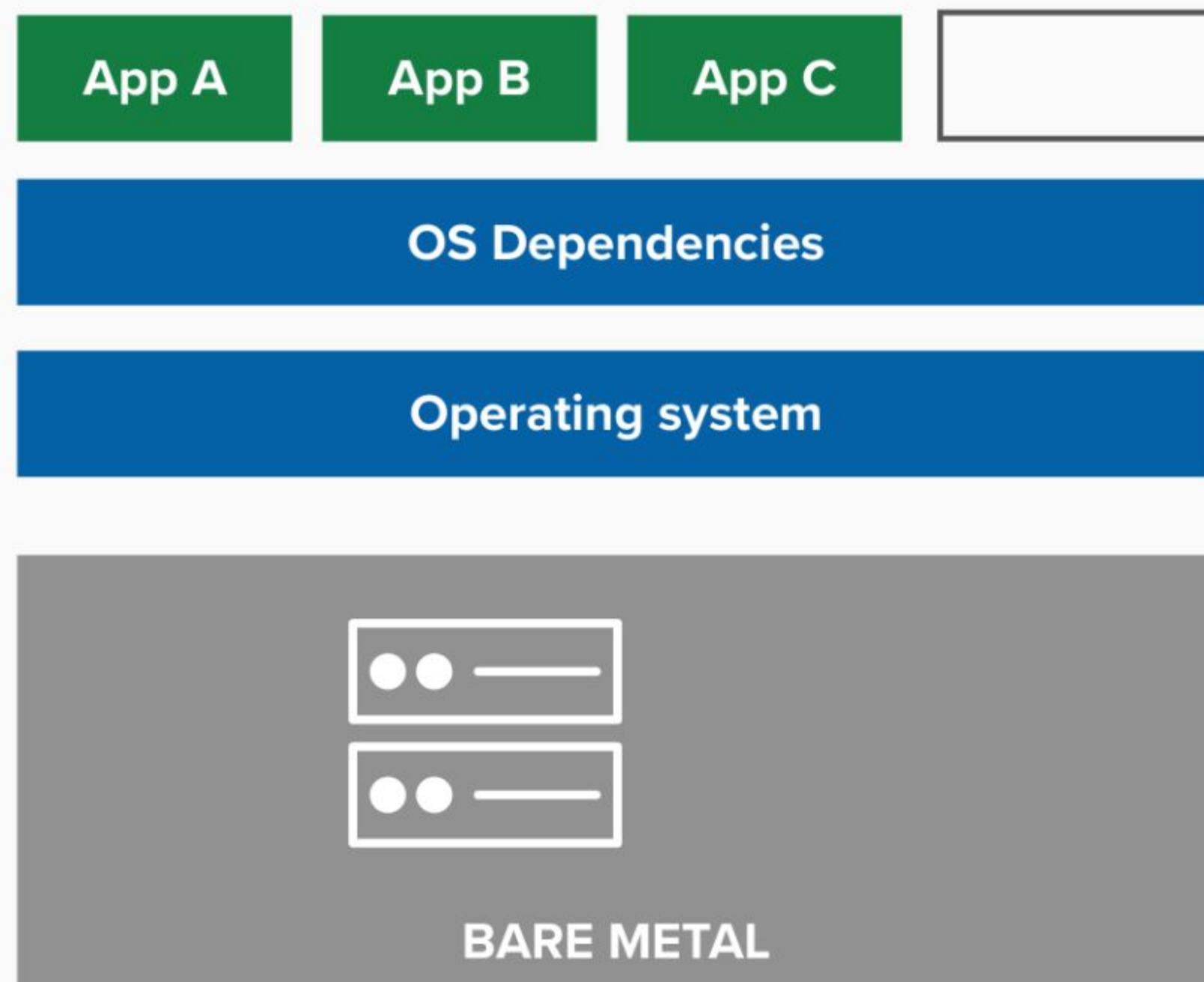




# 1. Scalability Challenge

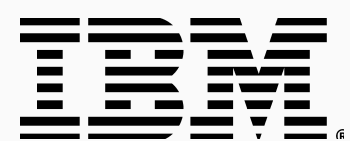


# 1. Vertical scaling



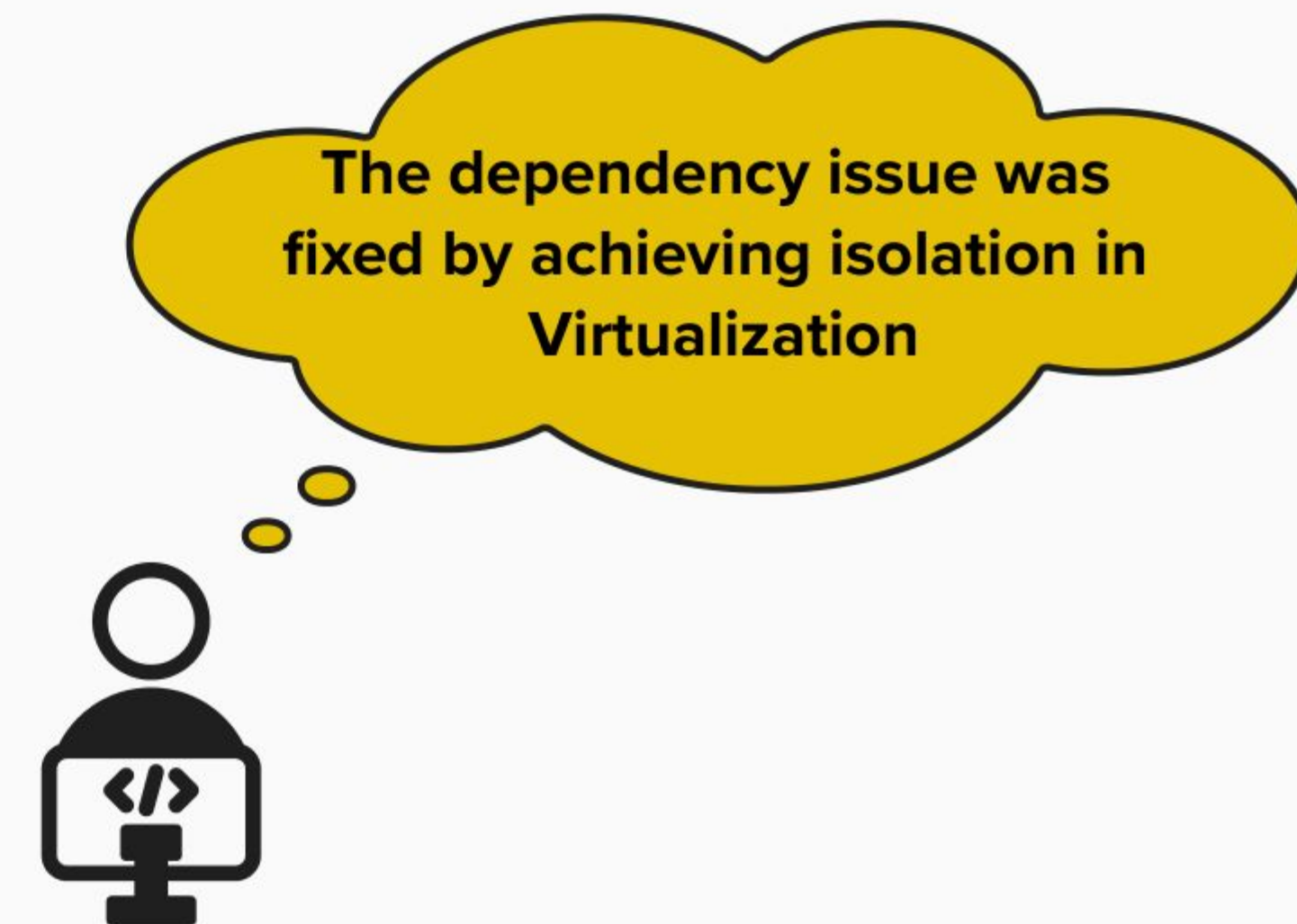
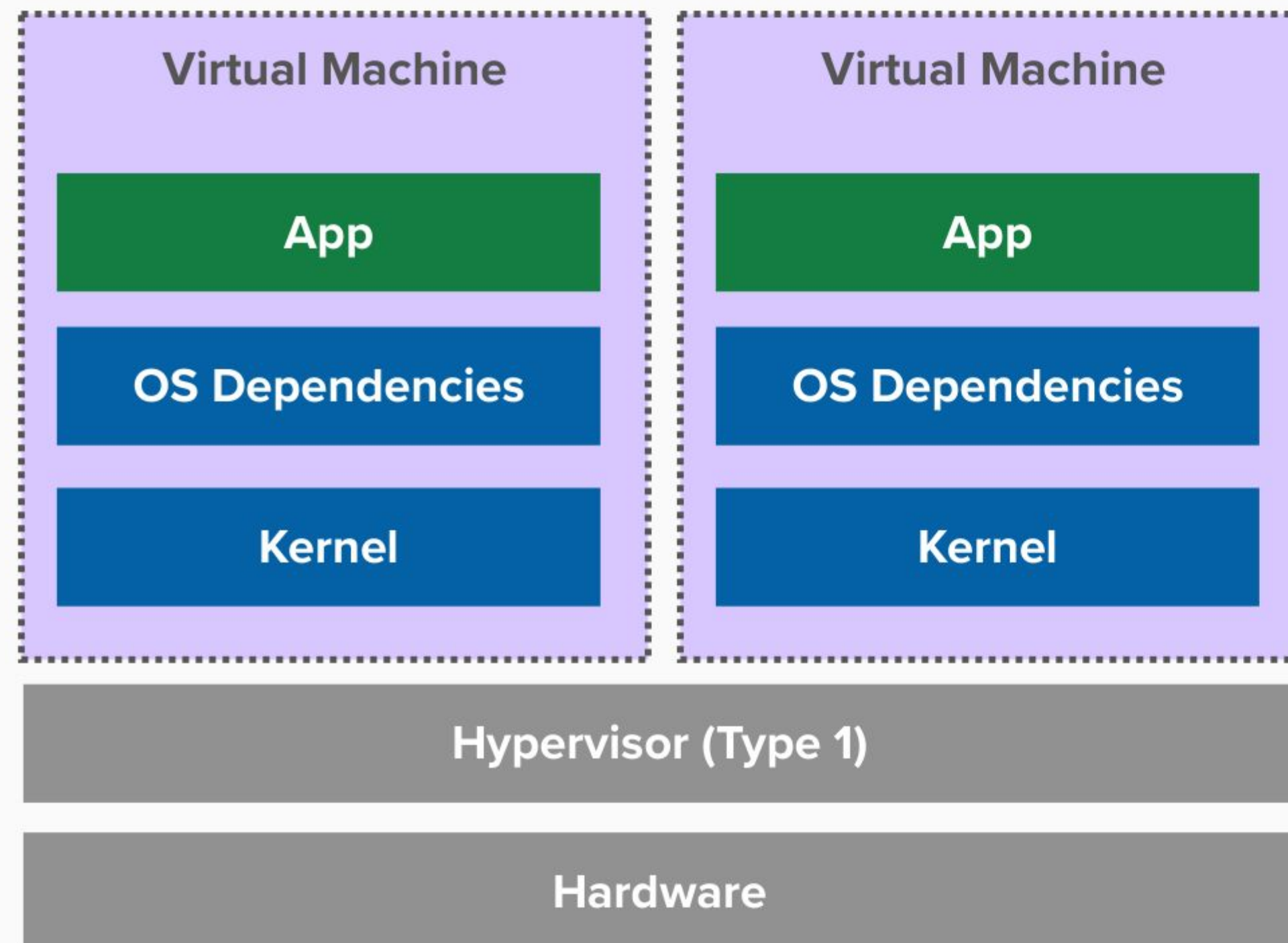
# 1. Vertical scaling problem

Once we run out of resources again, we will need to keep adding more and more resources to the host. This approach is neither scalable nor maintainable.



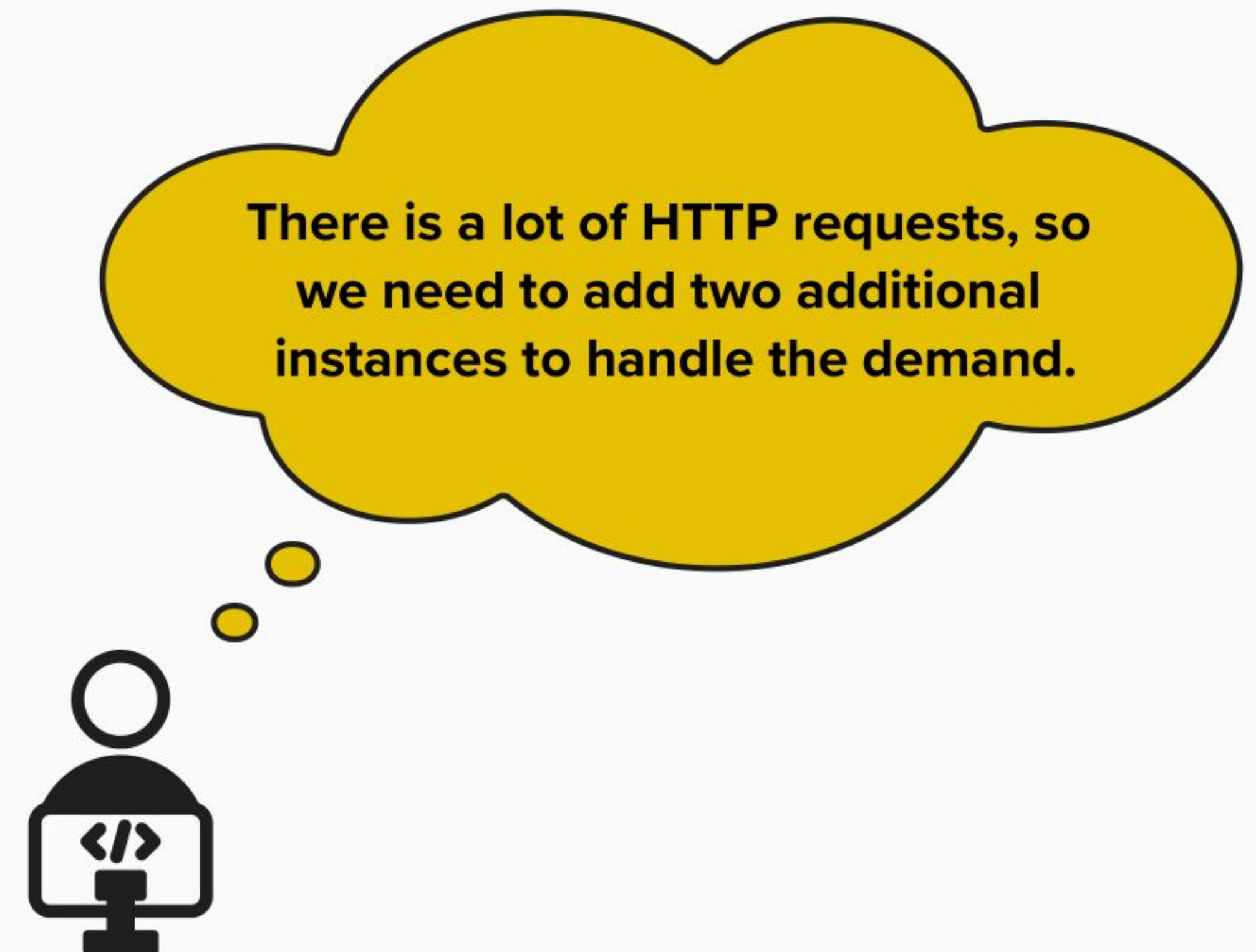
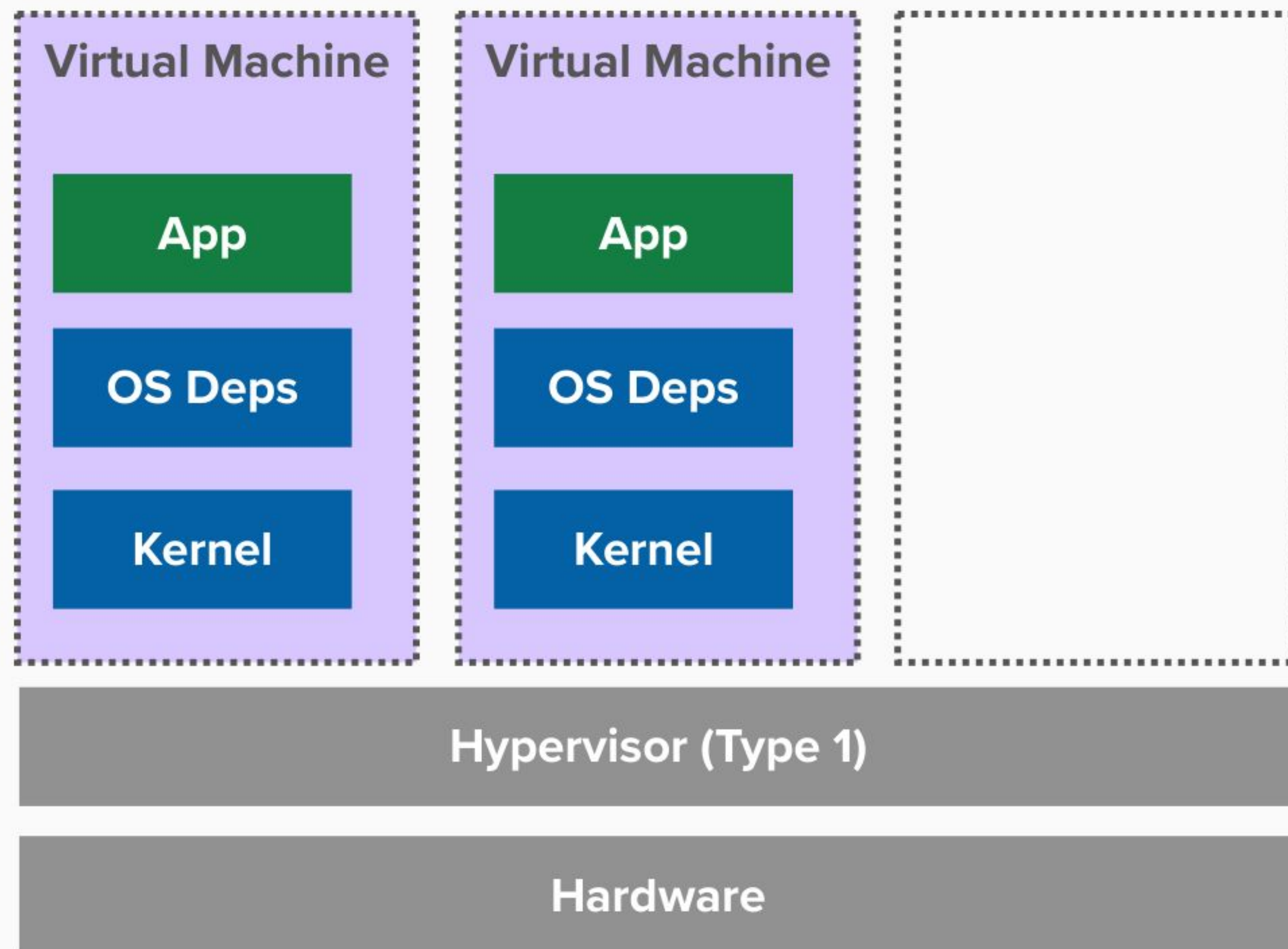


# 1. Virtualization as a Solution for Dependency Management Issues

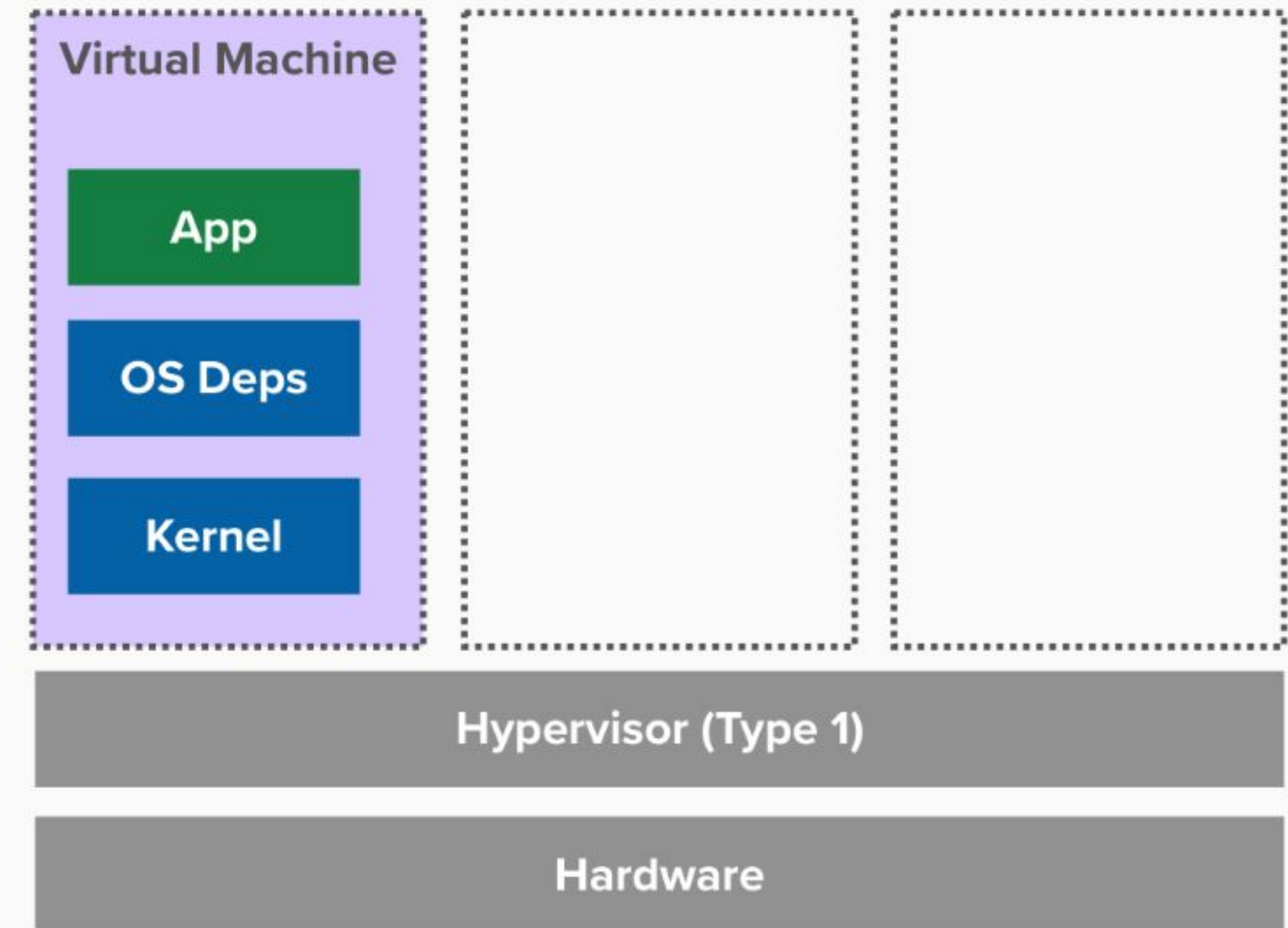
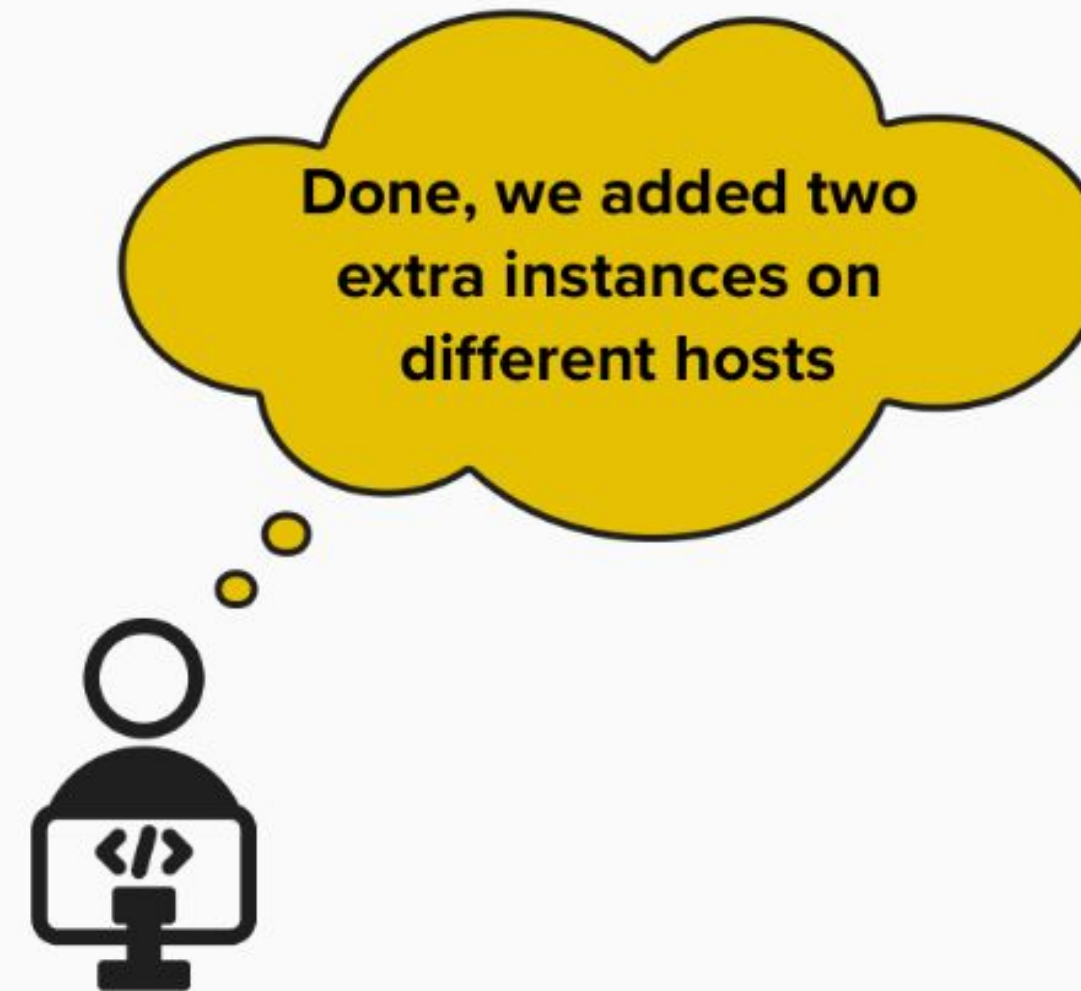
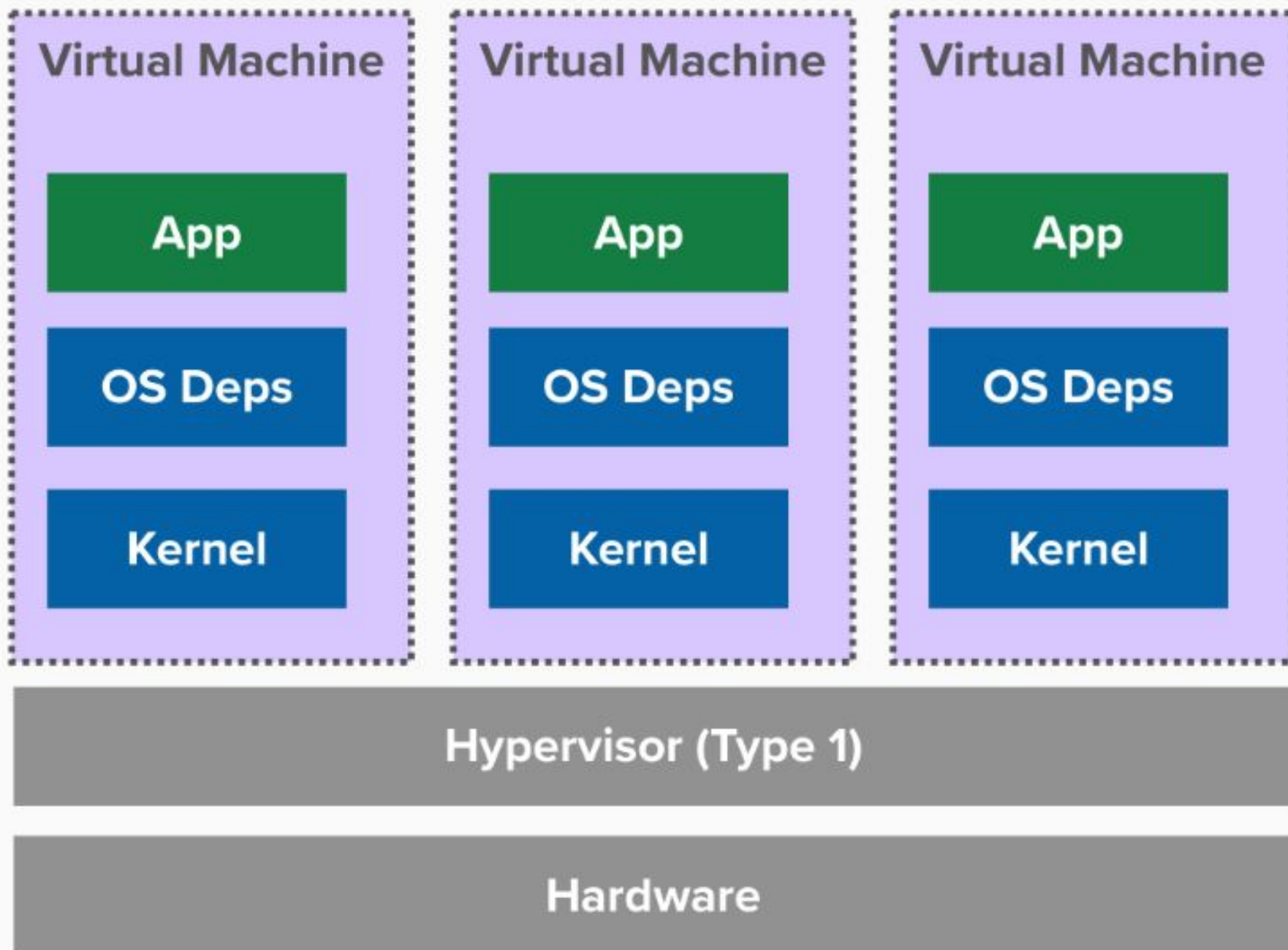




# 1. Virtualization as a Solution for scalability

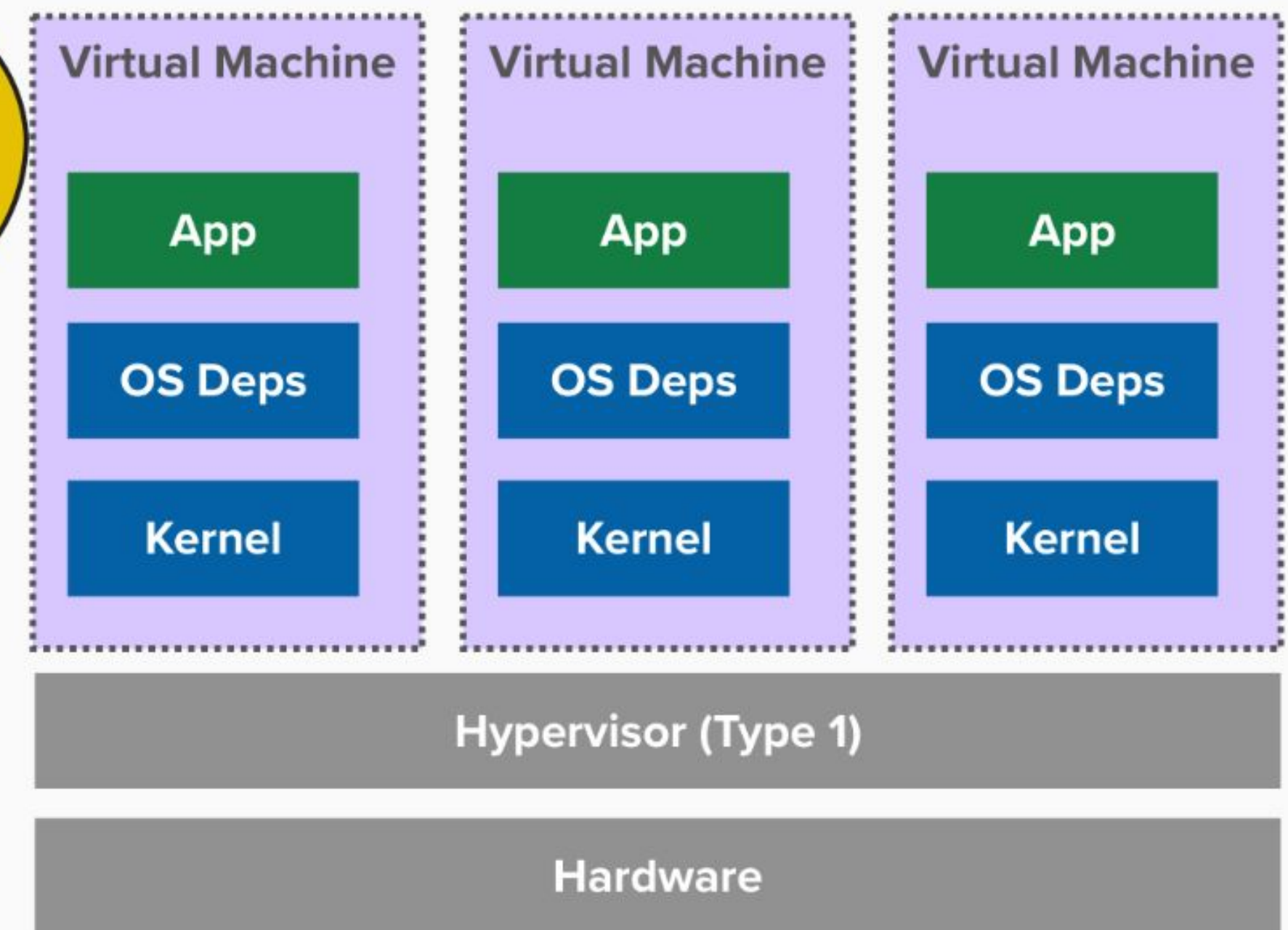
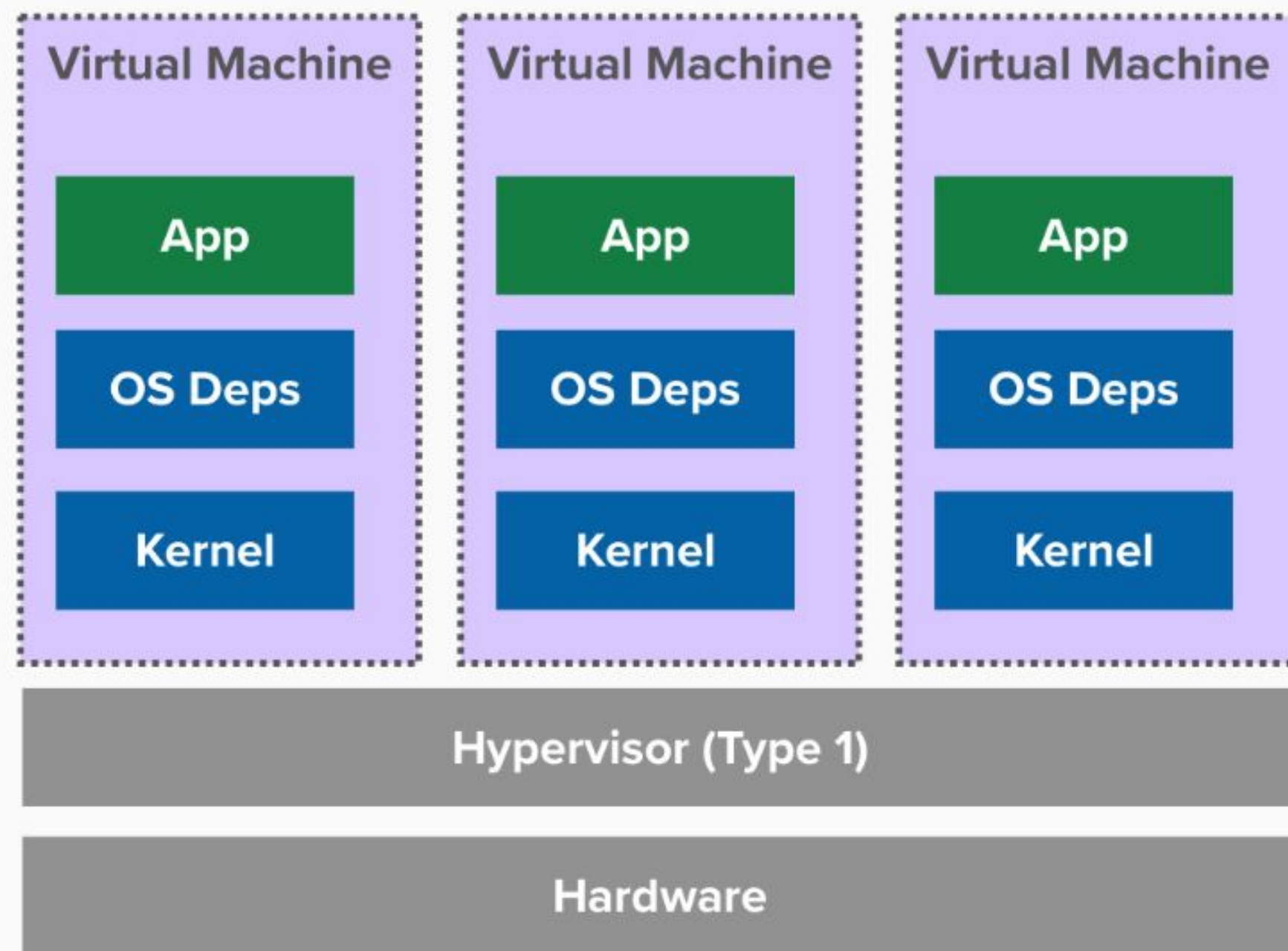


# 1. Virtualization as a solution for scalability



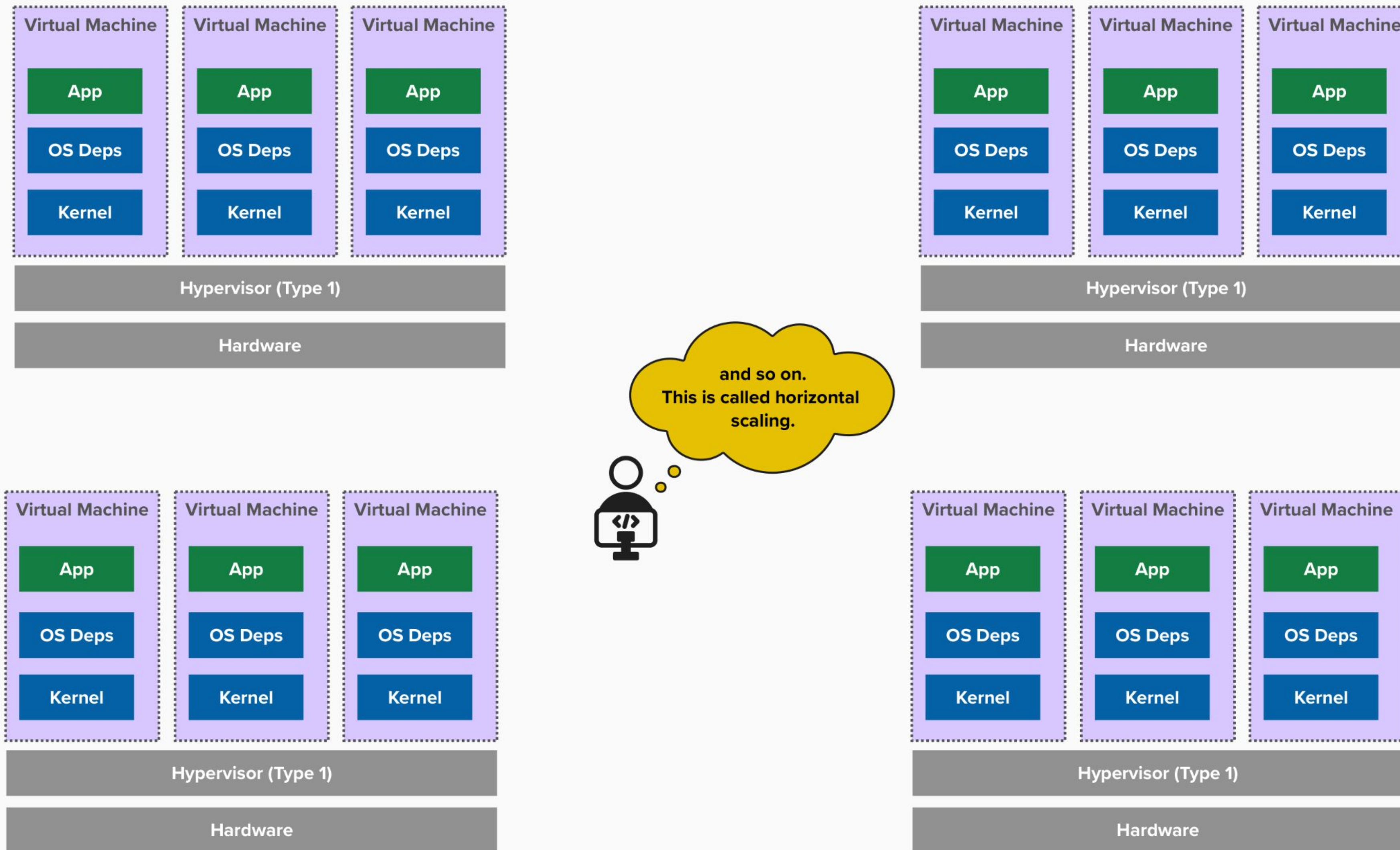


# 1. Virtualization as a solution for scalability





# 1. Virtualization as a solution for scalability

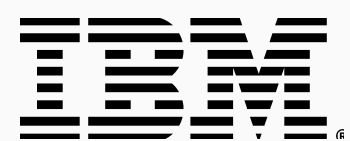




# 1. Virtualization as a Solution for scalability

However, there are some challenges:

- How can we deploy additional application instances more quickly and reduce VM startup time?
- How can we ensure application portability when VMs are migrated between different hosts?
- How can we package application dependencies effectively to make VMs more lightweight?



# CONTAINERIZATION AS A SOLUTION



**Red Hat**



## 2. What are containers?

- a **Set of Processes** sharing a common Kernel's resources for execution.
- **Isolated** from the rest of the machine.  
(can't see/affect/harm host or other containers)
- **Implementation:**
  - Uses **namespaces** to view the system's privately  
(network interfaces, PID tree, etc.)
  - Uses **cgroups** to have metered/limited/reserved resources.



## 2. What are containers?

- **Fast**

- Boots in milliseconds

- **Lightweight**

- Just a few MBs of intrinsic disk/memory usage

- **Portable**

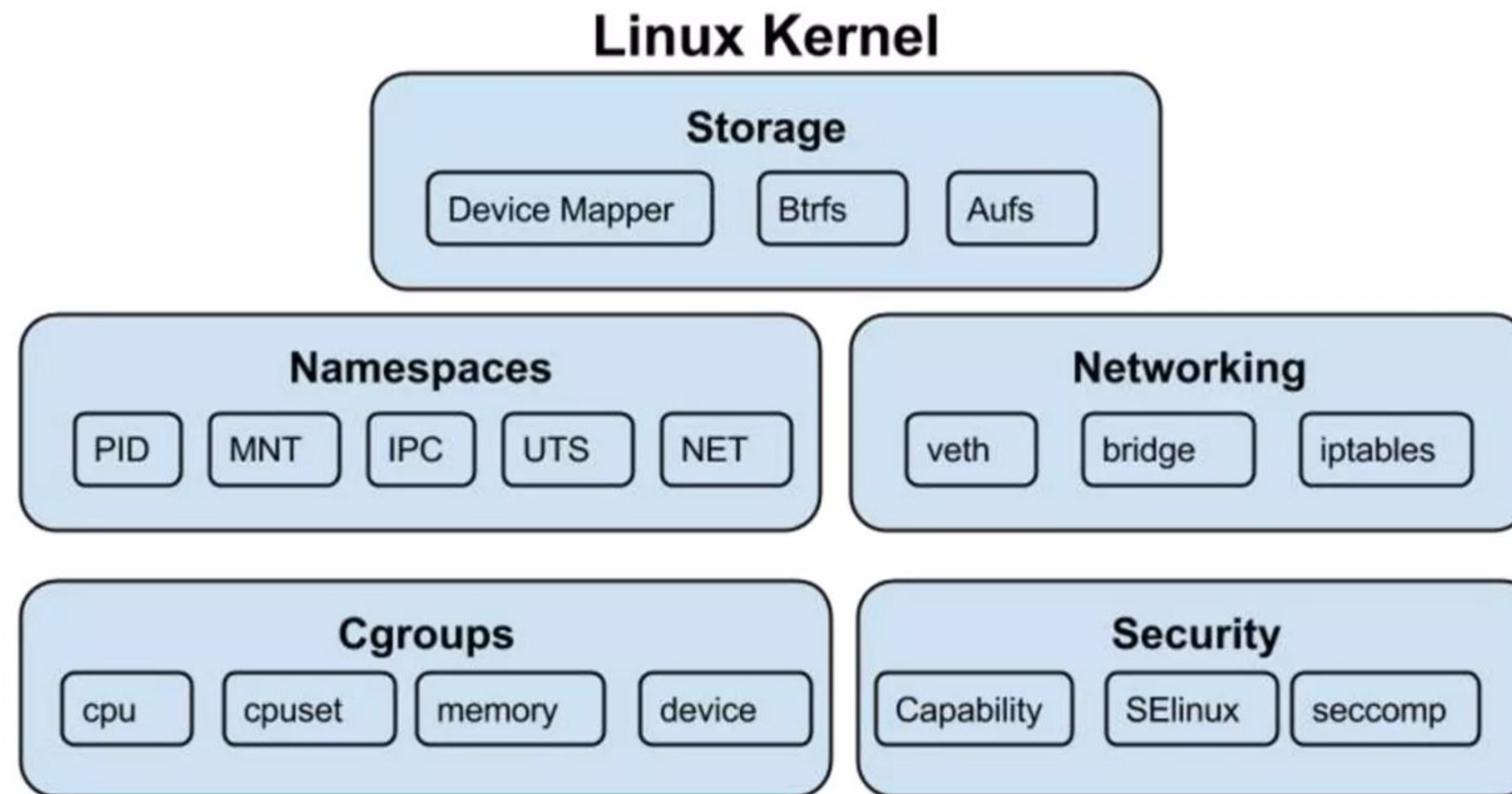
- Migration from one system to another like a VM.
- Deploy new applications or kill the old ones instantly.

- **Secure**

- Considerable level of security to applications, as Containers remain isolated from each other on the guest OS.



# Under the hood of Linux Containers



## 2. What are containers?

It depends who you ask

### Infrastructure

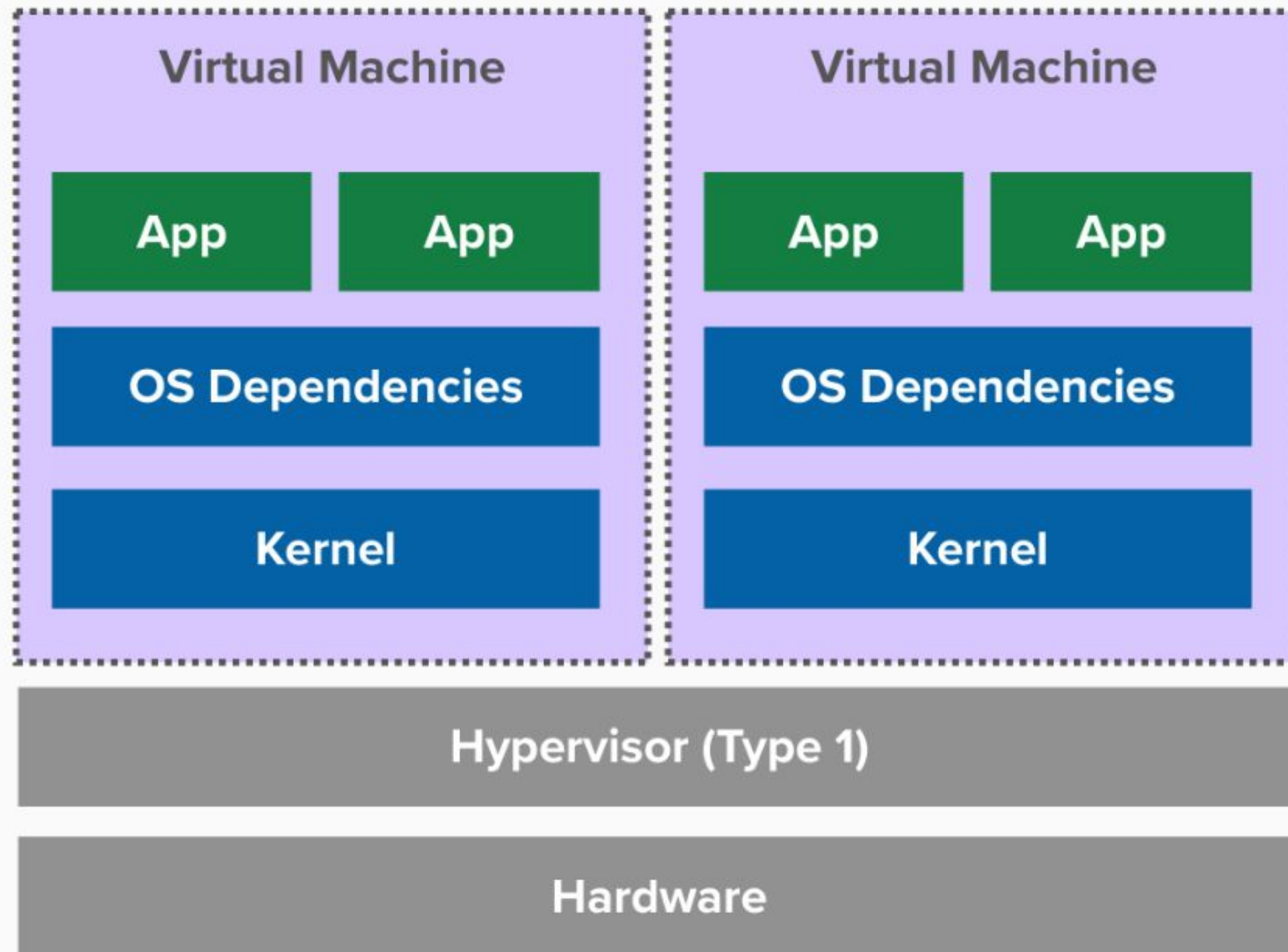
- Applications process on a shared kernel
- Simpler, lighter, and denser than VMS
- Portable across different environments

### Application

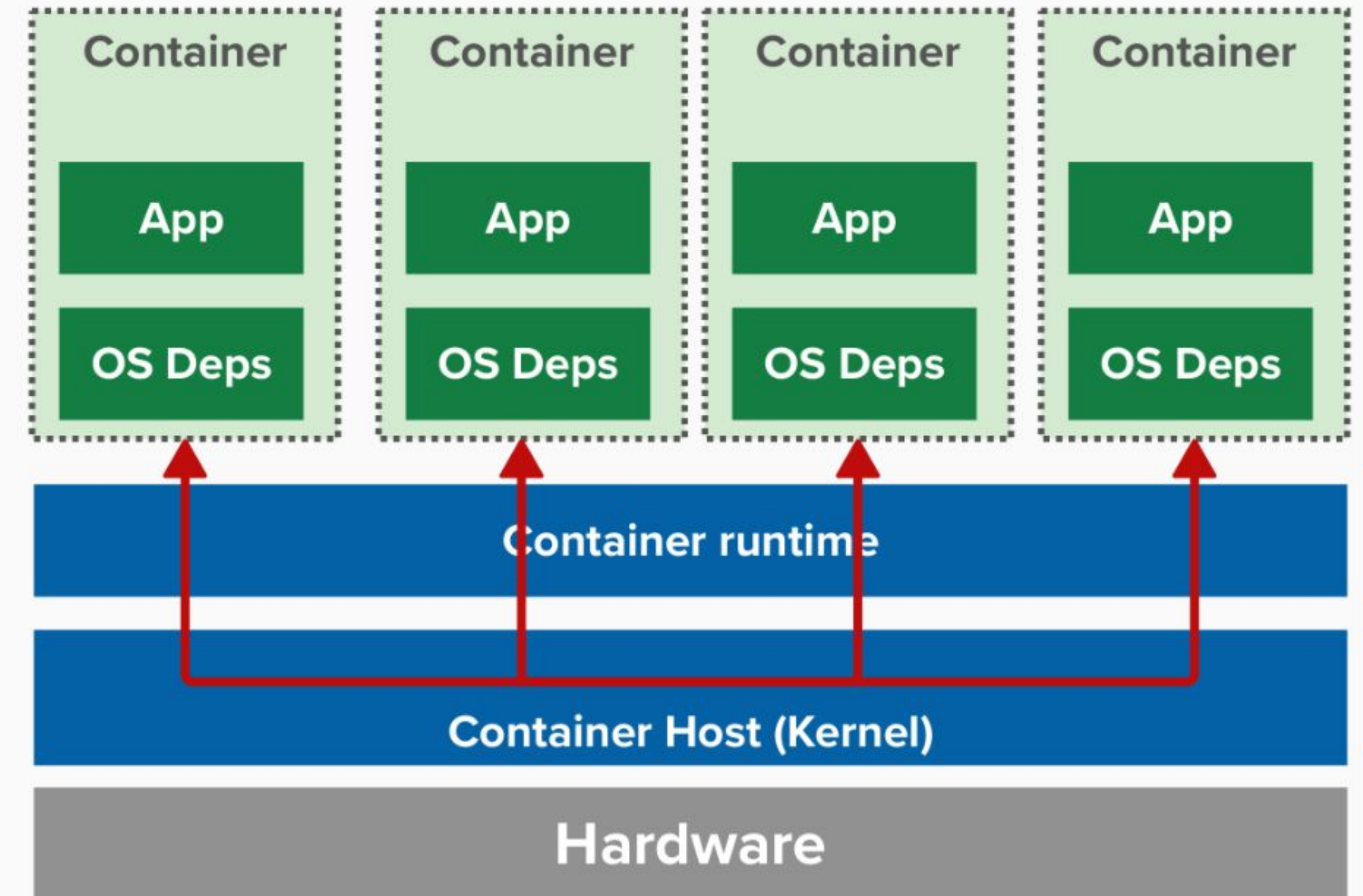
- Package apps with all dependencies
- Deploy to any environments in seconds
- Easily accessed and shared



## 2. Faster than a virtual Machine

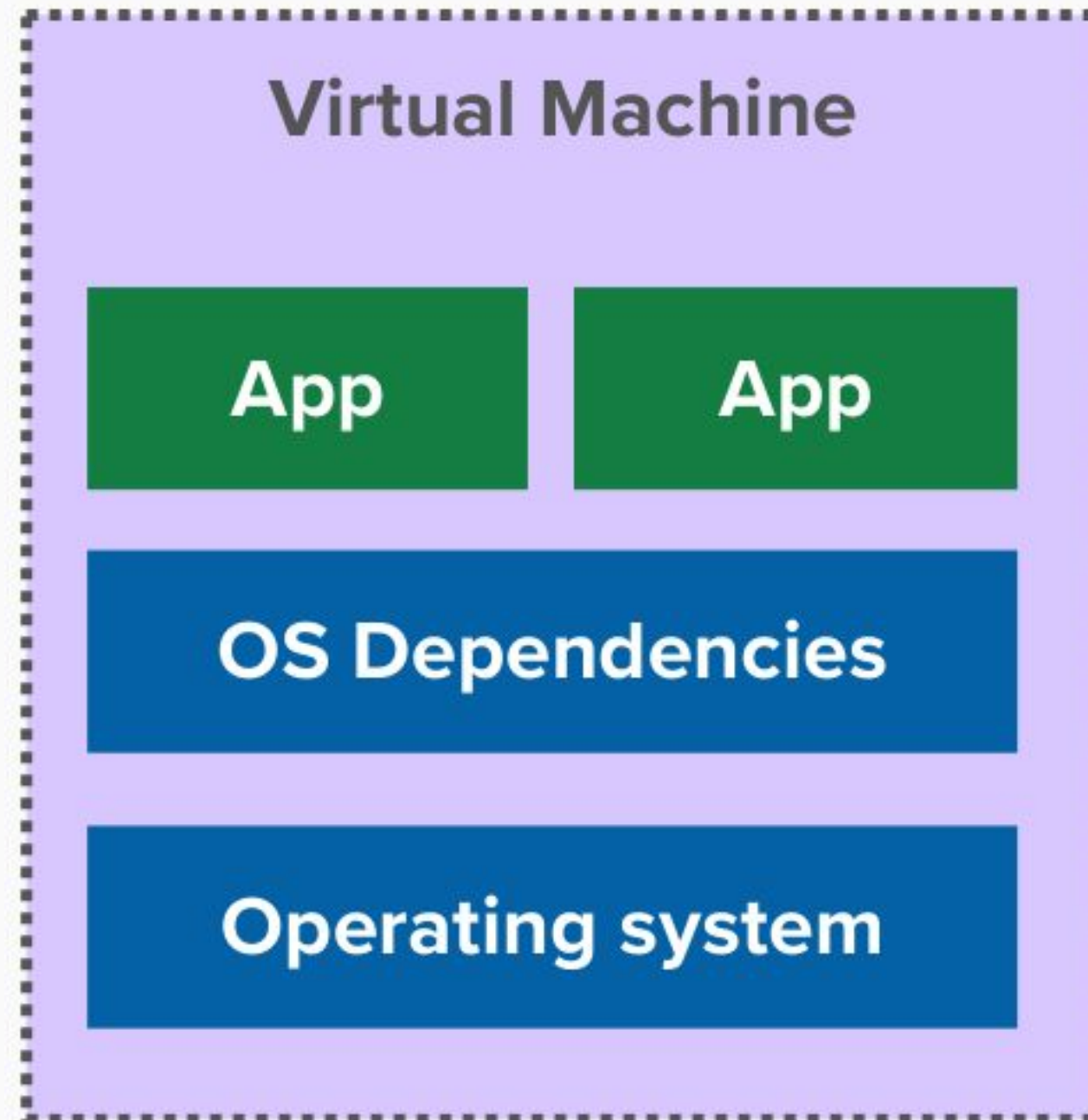


- Virtual machines are isolated
- App are not

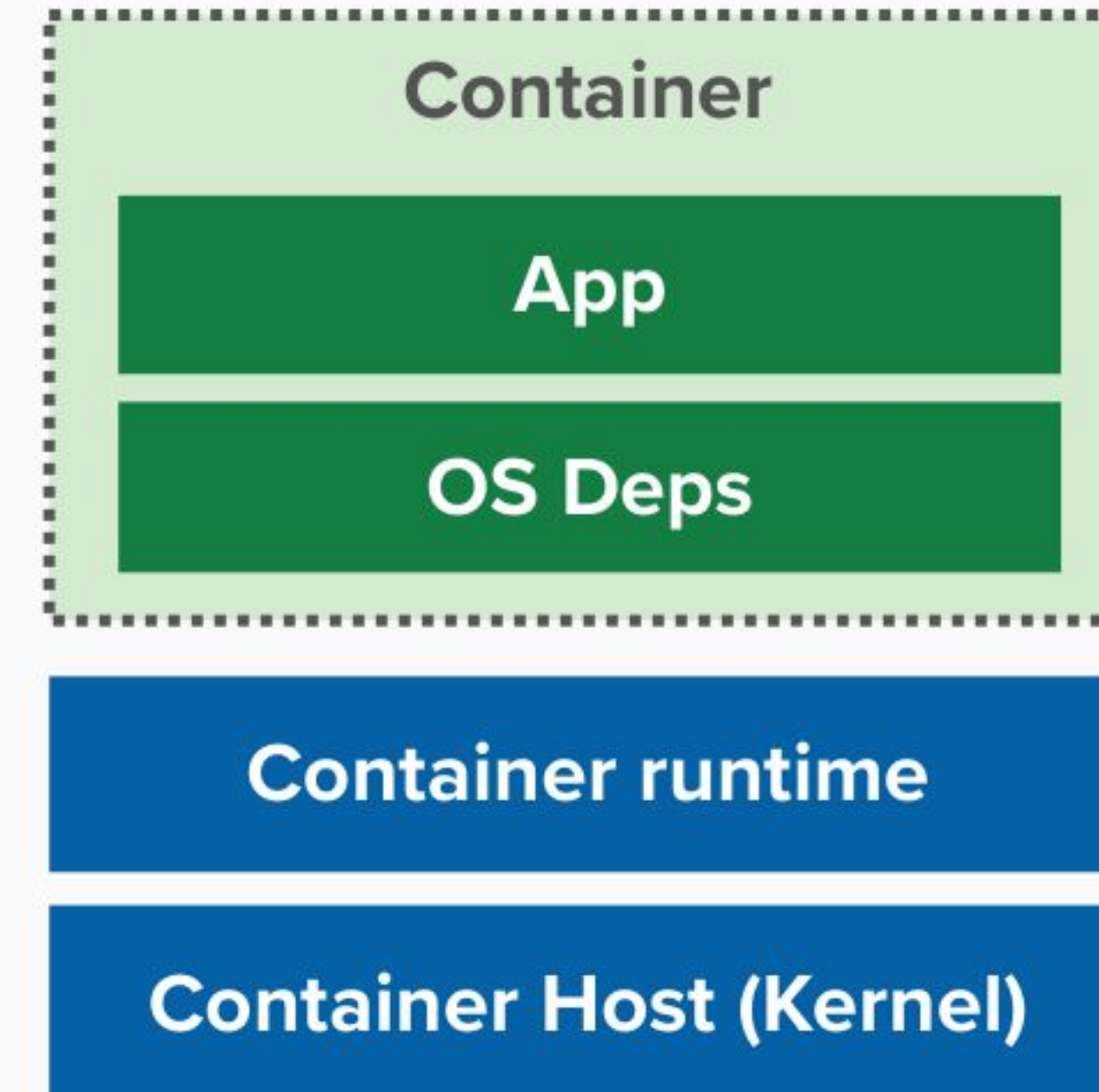


- Containers are isolated
- App are as well

## 2. Virtual Machines and Containers



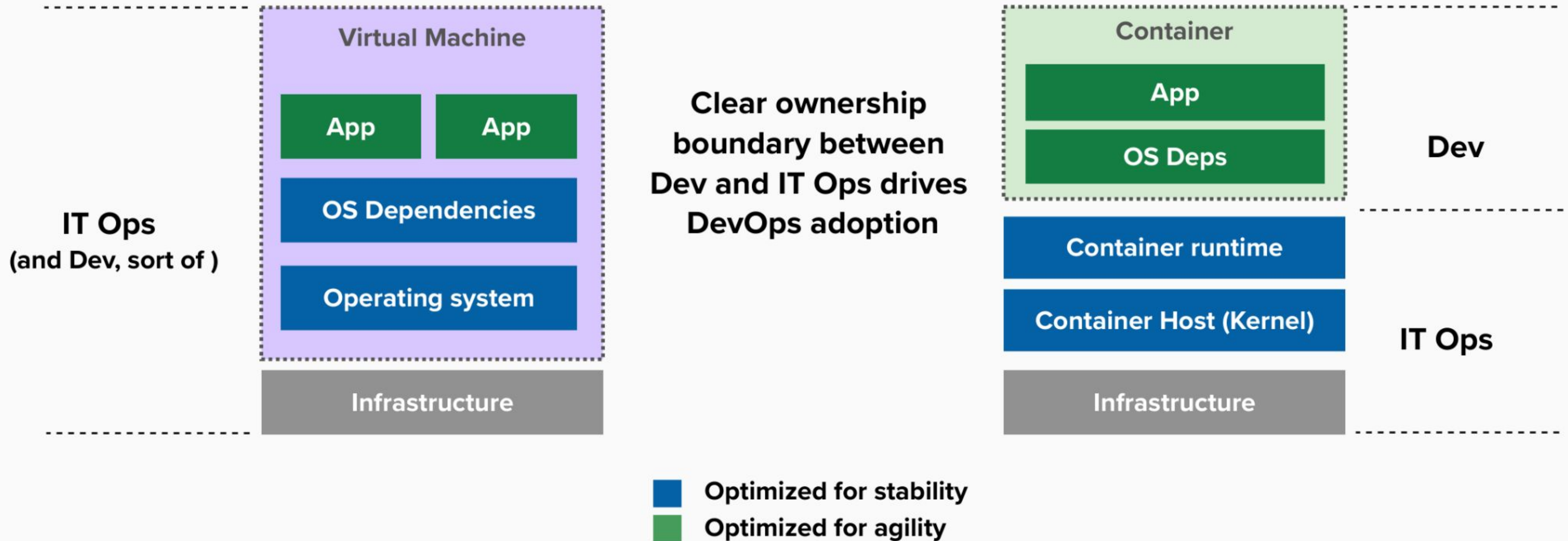
- + VM Isolation
- Complete OS
- Static Compute
- Static Memory
- High Resource Usage



- + Container Isolation
- + Shared kernel
- + Burstable Compute
- + Burstable Memory
- + Low Resource Usage



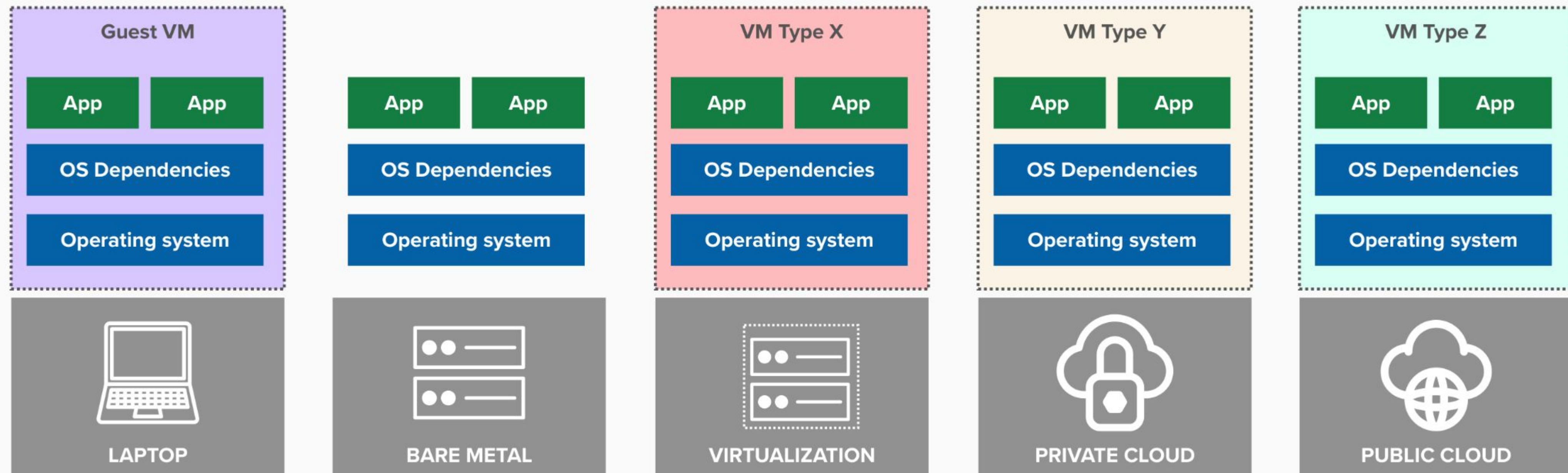
## 2. Virtual Machines and Containers





## 2. Application Portability with VM

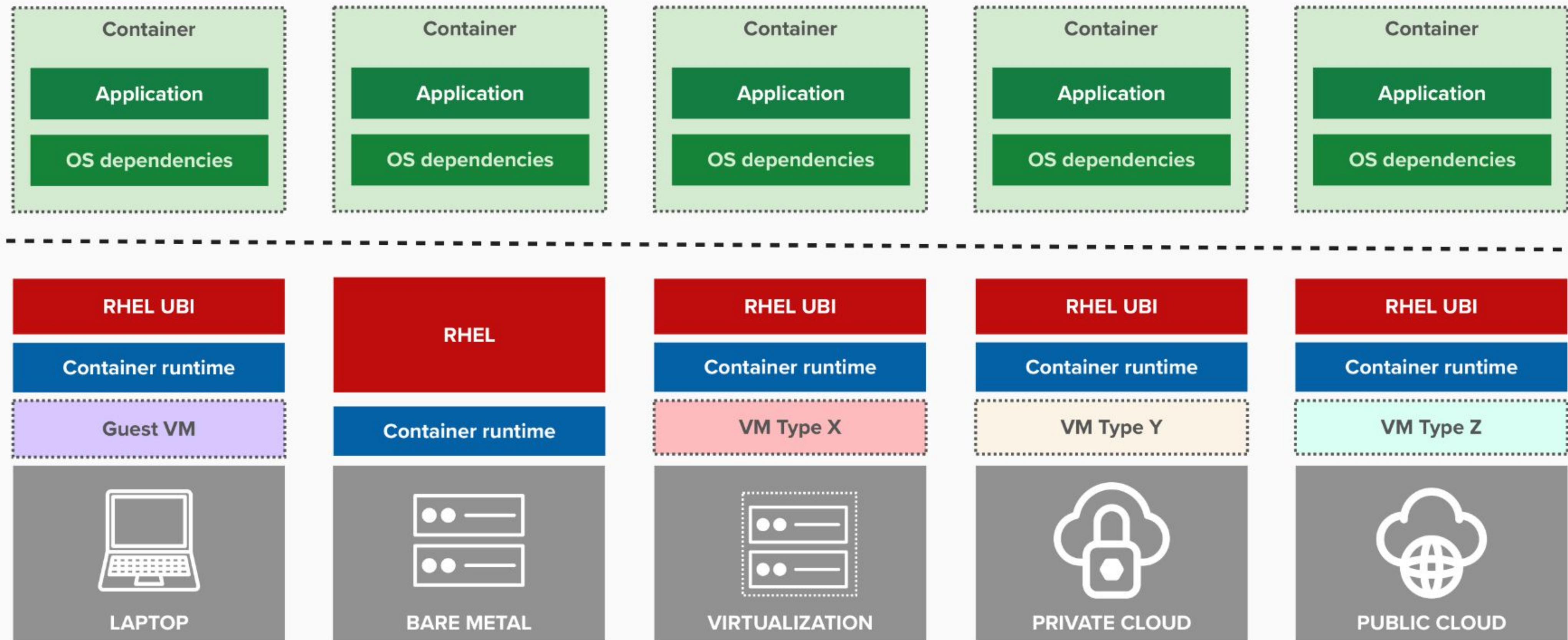
Virtual Machines are **NOT** portable across hypervisors and do **NOT** provide portable packaging for applications





# 2. Application Portability with Containers

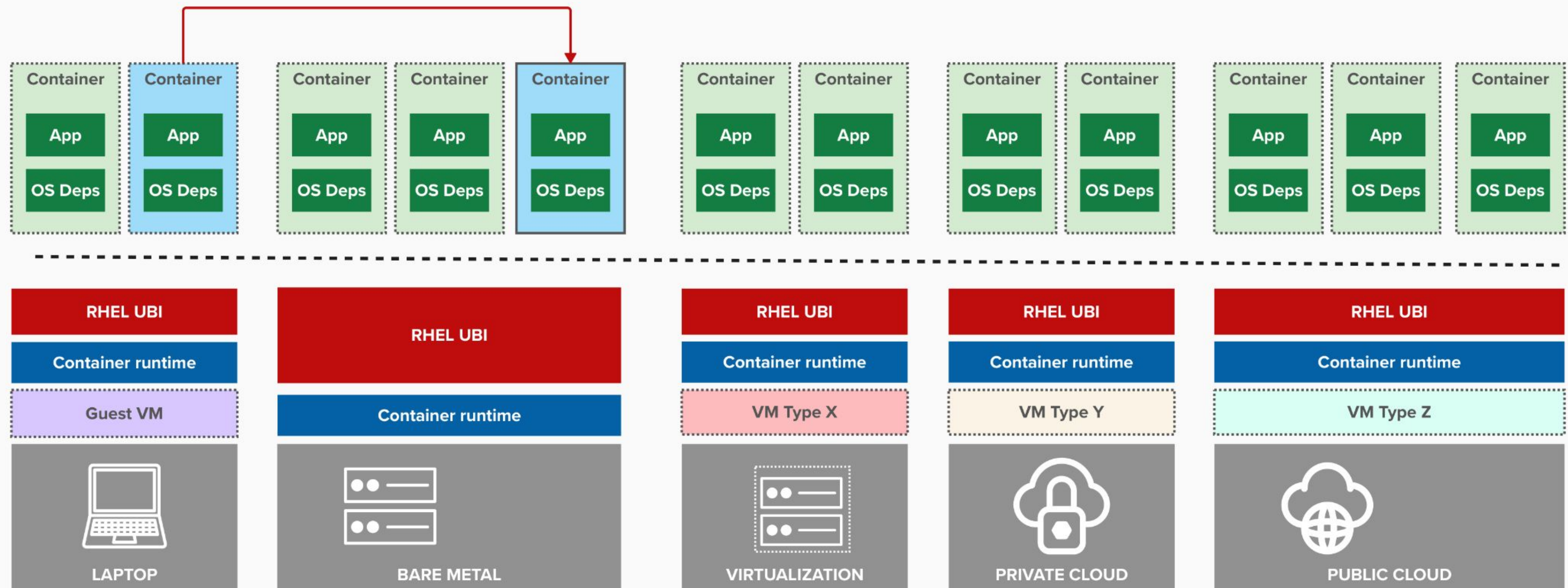
RHEL Containers + RHEL Host = Guaranteed Portability Across Any Infrastructure





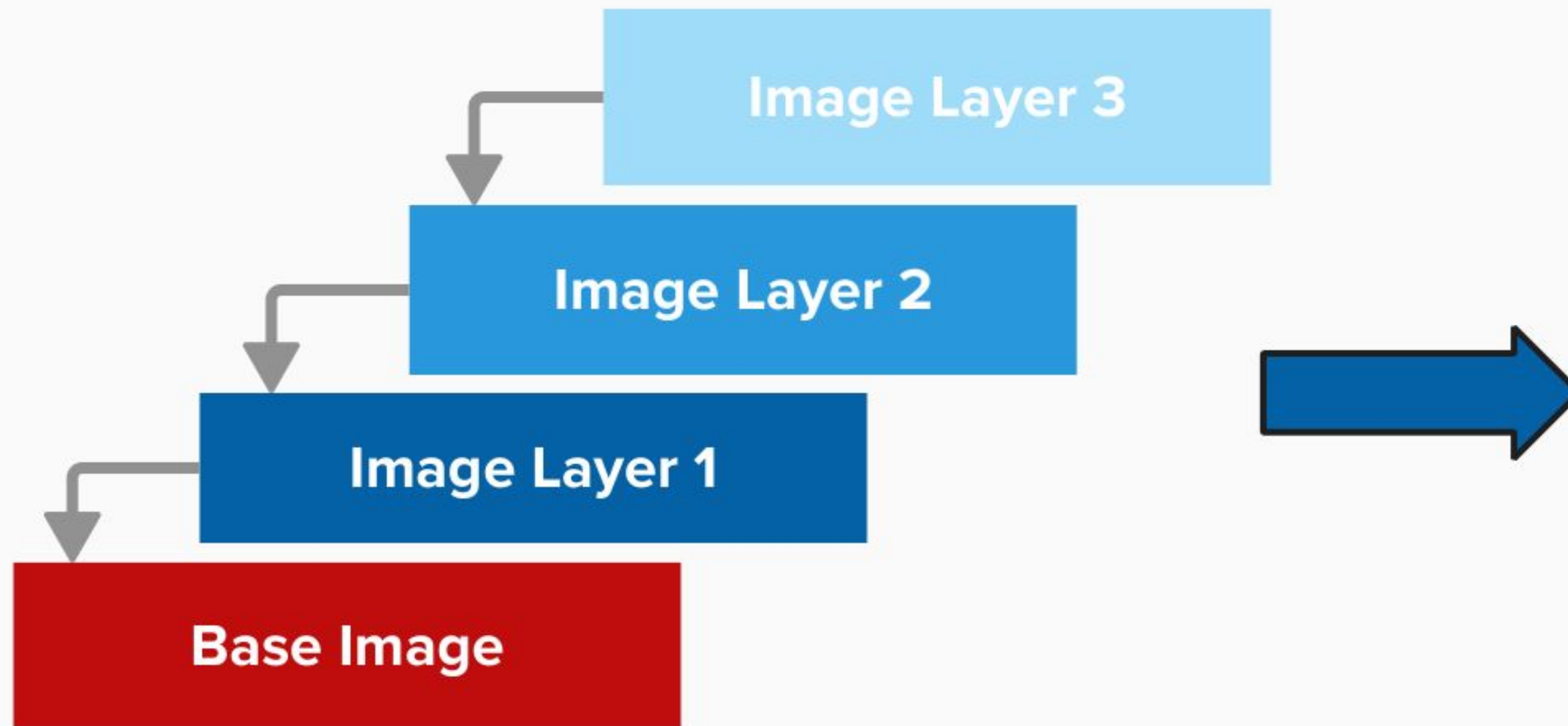
# 2. Application Portability with Containers

RHEL Containers + RHEL Host = Guaranteed Portability Across Any Infrastructure

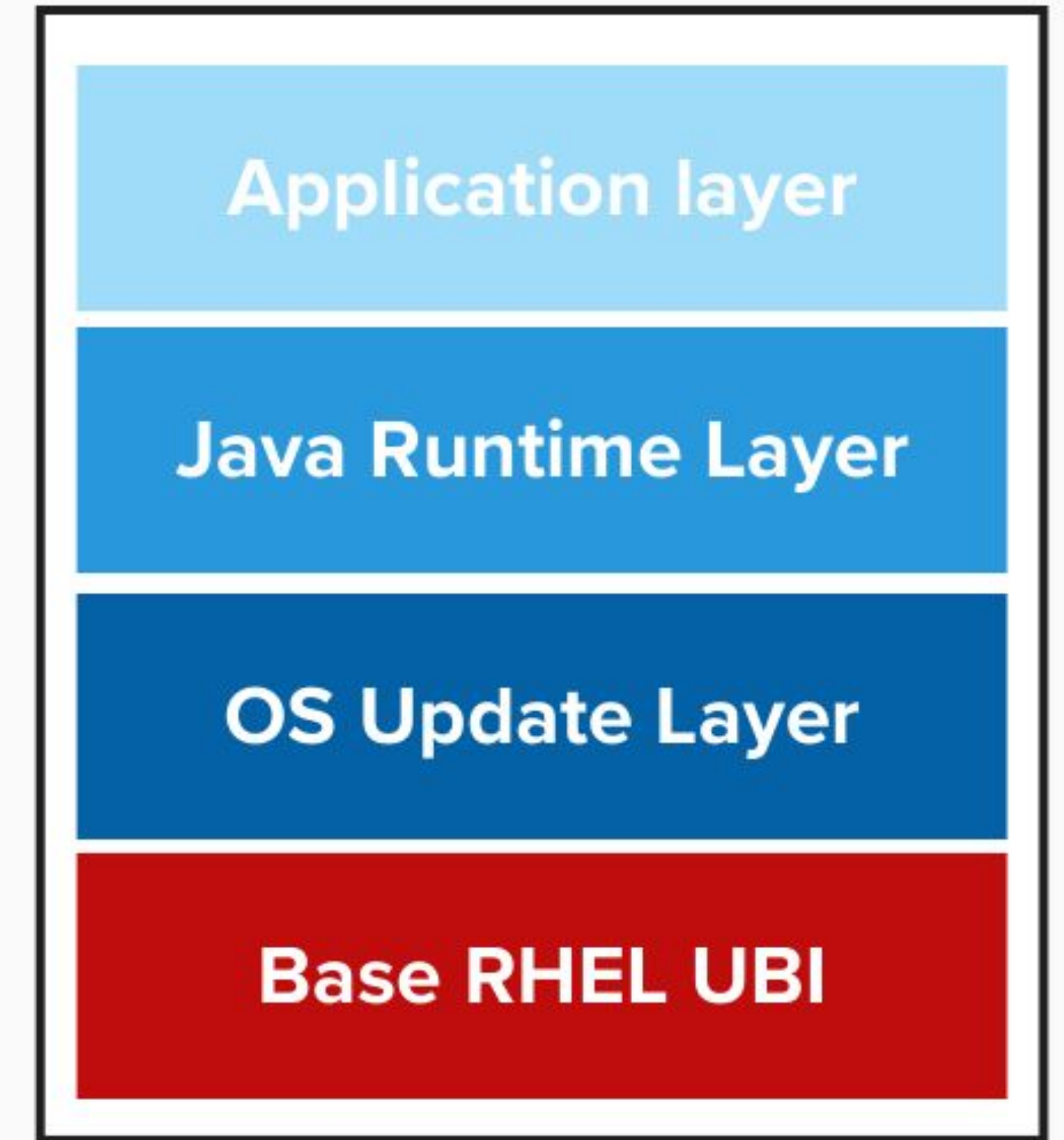




## 2. Rapid Security Patching Using Container Image Layering

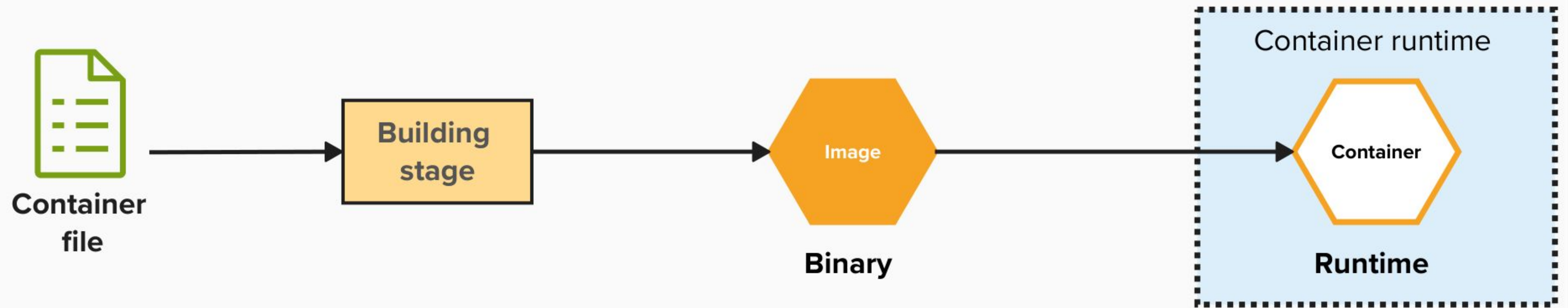


Container Image Layers



Container Image Layers

## 2. Containers are created from container images



# CONTAINER IMAGE REGISTRY

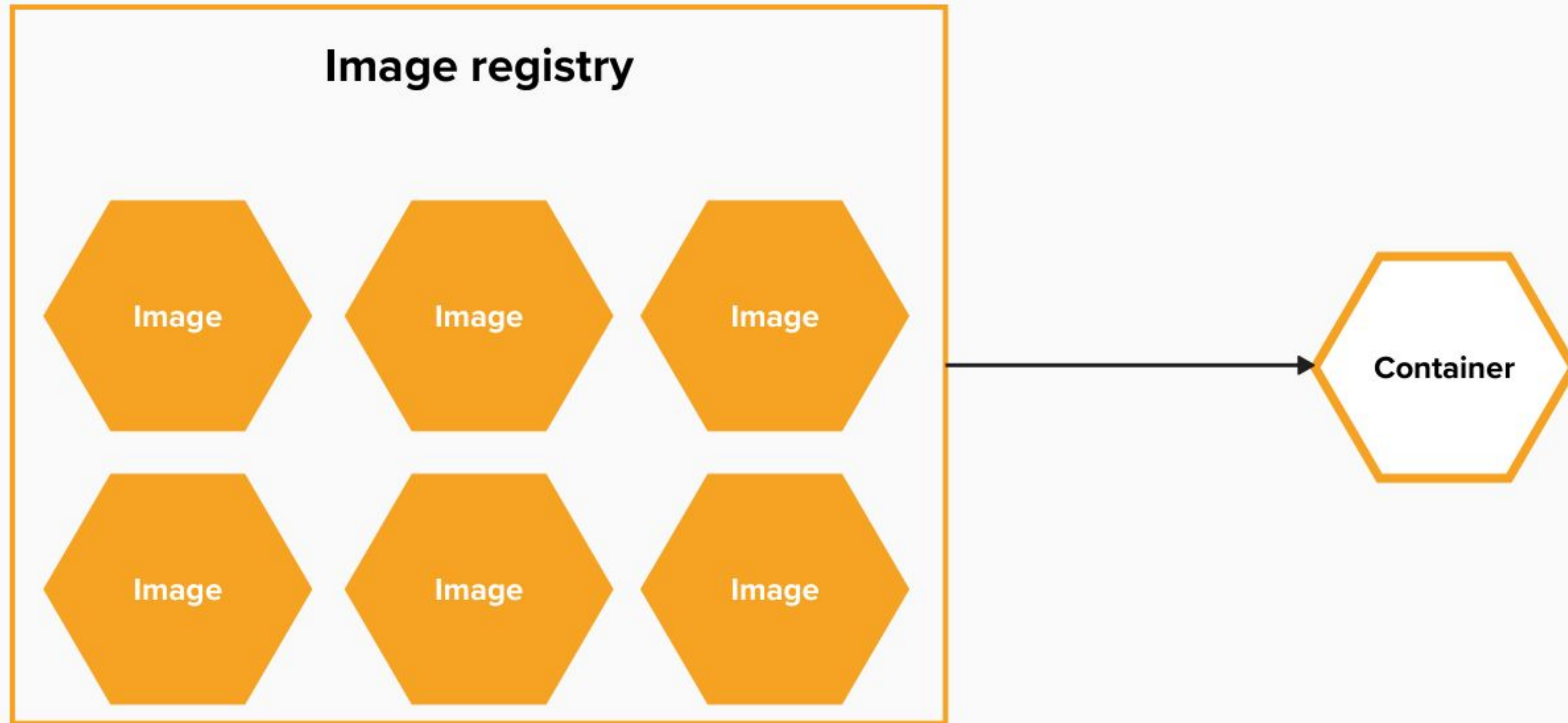


**Red Hat**

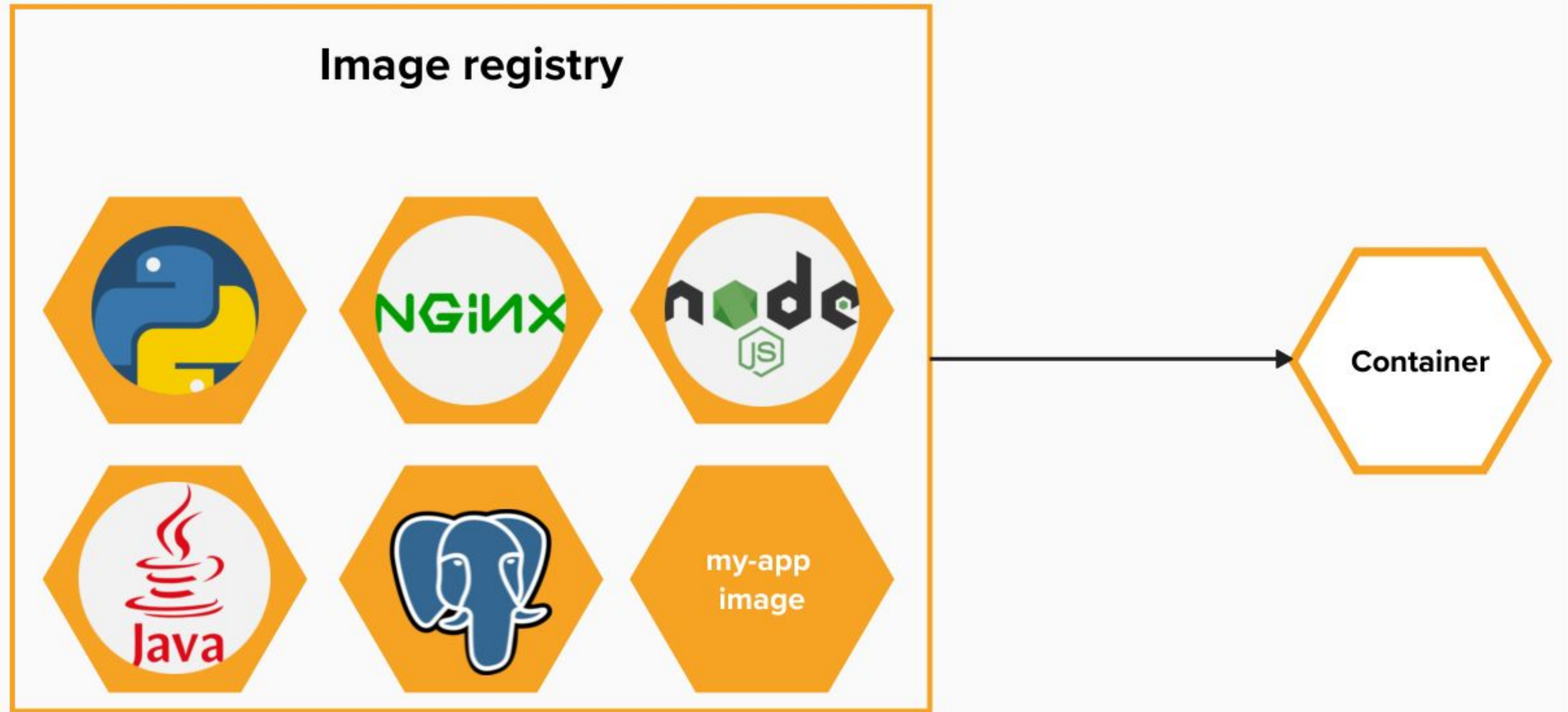




# Container images are stored in an image registry

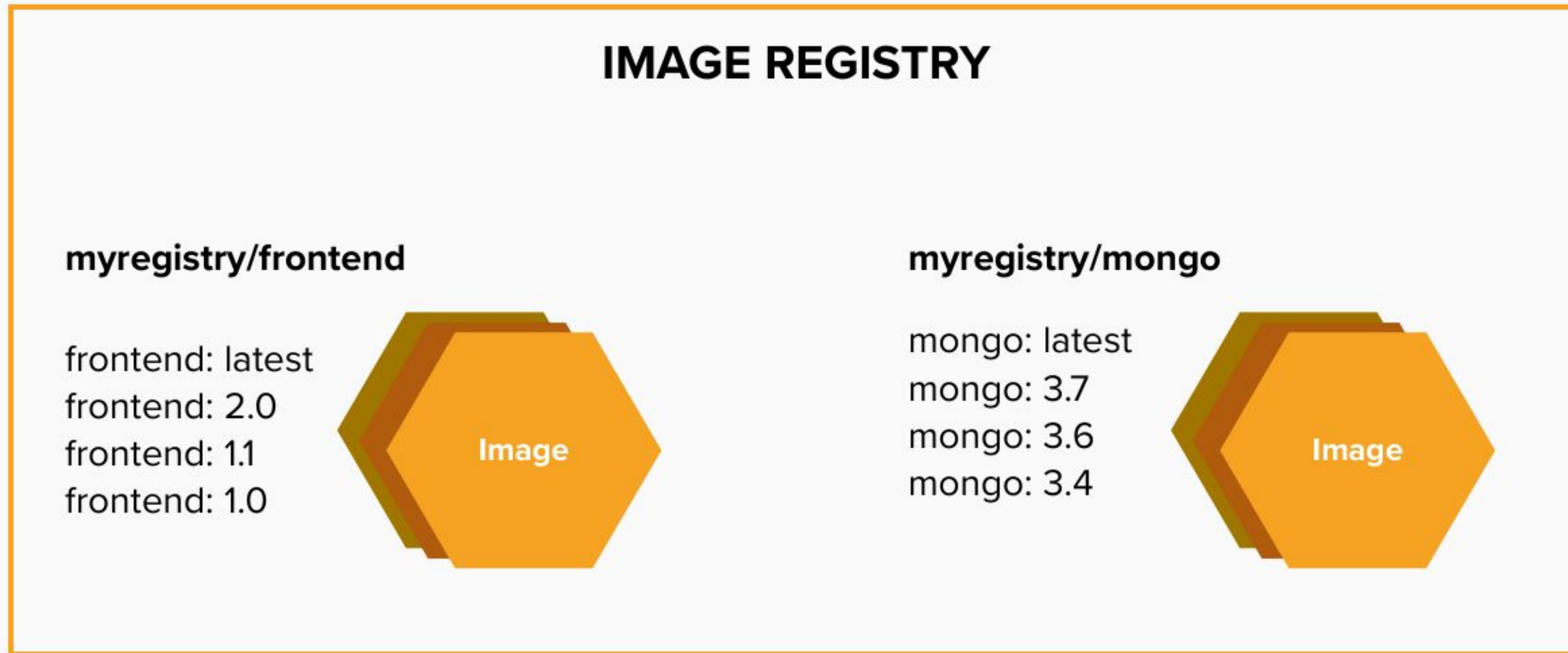


# Container images are stored in an image registry





# An image repository contains all versions of an image in the image registry





# Questions